

Anti-Derivatives & Inverse Trig

Section 5.7

1-7-26

A) Integrate

Ex: $\int \frac{1}{9+x^2} dx =$

$$\frac{1}{9} \int \frac{1}{1 + \frac{x^2}{9}} dx = \frac{1}{9} \int \frac{1}{1 + \left(\frac{x}{3}\right)^2} dx$$

$$= \frac{1}{3} \int \frac{1}{1+u^2} du = \frac{1}{3} \arctan(u) + C$$

$$= \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C$$

$u = \frac{x}{3}$
 $du = \frac{1}{3} dx$

INVERSE TRIG RULE 1!

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan \frac{u}{a} + C$$

(via u-sub!)

Ex: $\int \frac{x}{121+x^4} dx =$

$$= \int \frac{x}{11^2 + (x^2)^2} dx$$

$$a=11$$

$$u=x^2$$

$$du=2x dx$$

$$= \left[\frac{1}{2} \cdot \frac{1}{11} \arctan\left(\frac{x^2}{11}\right) + C \right]$$

Ex: $\int \frac{\cos x}{4 + \sin^2 x} dx =$

$$a=2$$

$$u = \sin x$$

$$du = \cos x dx$$

$$= \left[\frac{1}{2} \arctan\left(\frac{\sin x}{2}\right) + C \right]$$

INVERSE TRIG RULE 2!

$$\int \frac{1}{\sqrt{(a)^2 - (u)^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

(via u-sub!)

Ex: $\int_0^1 \frac{1}{\sqrt{4-x^2}} dx =$

$$= \arcsin\left(\frac{x}{2}\right) \Big|_0^1$$

$$= \arcsin\left(\frac{1}{2}\right) - \arcsin(0)$$

$$= \boxed{\frac{\pi}{6}}$$

INVERSE TRIG RULE 3!

$$\int \frac{1}{u\sqrt{u^2-a^2}} du = \frac{1}{a} \operatorname{arcsec} \frac{|u|}{a} + C$$

(via u-sub!)

Ex: COMPARE:

$$\int \frac{e^{2x}}{4+2e^{2x}} dx \quad \text{VS} \quad \int \frac{e^x}{4+e^{2x}} dx$$

Regular U-Sub | Inverse Trig

$$u = e^{2x}$$

$$du = 2e^{2x} dx$$

$$u = e^x$$

$$a = 2$$

$$\frac{1}{2} \arctan\left(\frac{e^x}{2}\right) + C$$

B) Complete the Square

Ex: $\int \frac{1}{\sqrt{-x^2+2x}} dx =$

$$\begin{aligned} &-(x^2-2x) \\ &-(x^2-2x+1-1) \\ &-((x-1)^2-1) \end{aligned}$$

$$\int \frac{1}{\sqrt{1-(x-1)^2}} = \arcsin(x-1) + C$$

$$a=1$$

$$u=x-1$$

Ex: $\int \frac{x}{\sqrt{9-8x^2-x^4}} dx =$

$$\begin{aligned} &-(x^4+8x^2-9) \\ &-(x^4+8x^2+16-9-16) \\ &-((x^2+4)^2-25) \end{aligned}$$

$$\int \frac{x}{\sqrt{25-(x^2+4)^2}} = \left[\frac{1}{2} \arcsin\left(\frac{x^2+4}{5}\right) + C \right]$$

$$a=5$$

$$u=x^2+4$$

$$du=2x dx$$

Ex: $\int \frac{2x-5}{x^2+2x+2} dx =$

$$u = x^2+2x+2$$

$$du = 2x+2 dx$$

doesn't match numerator, so we MAKE IT

$$\int \frac{2x+2-5-2}{x^2+2x+2} dx = \int \frac{2x+2}{x^2+2x+2} dx + \int \frac{-7}{x^2+2x+2} dx$$

$$u = x^2+2x+2$$

$$du = 2x+2 dx$$

$$\int \frac{1}{u} du = \ln|u| = \ln|x^2+2x+2|$$

$$\int \frac{-7}{x^2+2x+2} dx =$$

$$\hookrightarrow (x^2+2x+2)$$

$$(x^2+2x+1+2-1)$$

$$(x+1)^2+1$$

$$\int \frac{-7}{((x+1)^2+1)} dx = -7 \arctan(x+1) + C$$

$$a=1$$

$$u=x+1$$

$$\rightarrow \ln|x^2+2x+2| - 7 \arctan(x+1) + C$$